Data for calculating the beam profile of THz beam produced via two-color filamentation in air

UNIS group

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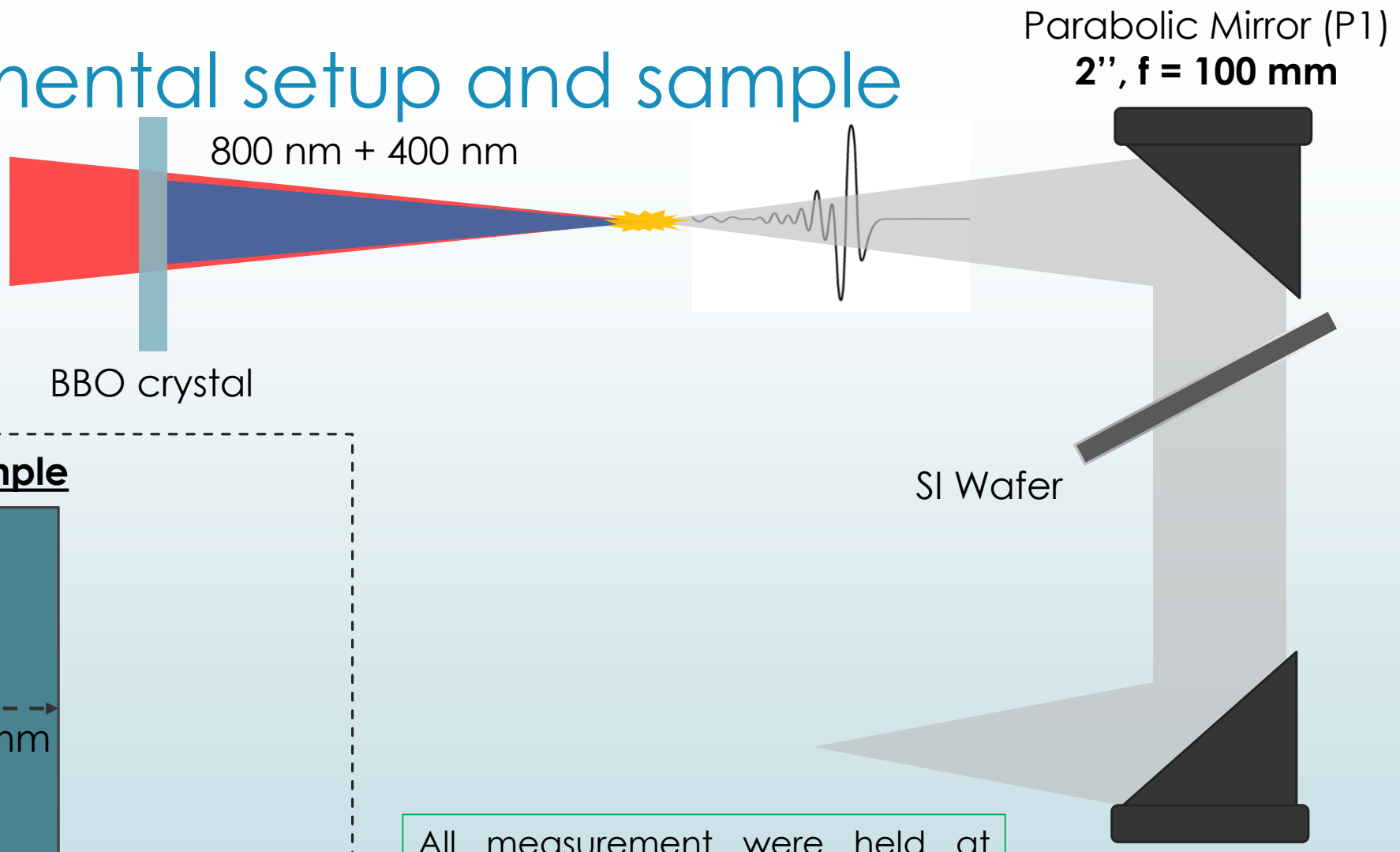
Experimental setup and sample



Sample

3 mm thick HDPE
(High Density Polyethylene sheet)

All measurement were held at almost the same relative humidity:
RH = 2.9 %

Information about the experimental setup

- The sample is a 3 mm thick High Density Polyethylene sheet (HDPE)
- The parabolic mirror (P1) used to collect the produced THz beam is: **2'' Ø , f = 100 mm**
- The focusing parabolic mirror (P2) is: **2'' Ø , f = 150 mm**
- In order to block residual IR and visible light, a **2'' 1.5 mm thick Si wafer** is placed after the first parabolic mirror P1 at Brewster angle
- THz electric field produced via two-color filamentation in air is recorded using standard electro-optic detection technique. Specifically, all measurements have been performed using a **ZnTe** e.o. crystal of **500 μm** thickness, which exhibits a detection THz bandwidth from 0.4 THz to 2.68 THz .
- All measurements were held at almost the same relative humidity **RH = 2.9 %**

Experimental measurements 1

Measurements of the THz electric field of UNIS source

- **Reference:** Recorded THz electric field without the presence of any sample

  Files: Reference/Reference_00.dat, _01.dat, _02.dat

- **Sample (HDPE) is placed :**

- after the filament: Transmitted THz electric field through the sample, which is placed after the filament, before the first parabolic mirror (P1)

  Files: After filament/HDPE_after_filament_00.dat, _01.dat, _02.dat

- after P1(collimated THz beam): Transmitted THz electric field through the sample which is placed after the Si wafer and before the parabolic mirror (P2) where the THz beam is collimated

  Files: Collimated/HDPE_collimated_00.dat, _01.dat, _02.dat

Experimental measurements 2

Measurements of the THz electric field of UNIS source

(the THz focus is defined experimentally and that is the reason why we measured the THz transmission through the sample at three different places around the defined THz focus)

➨ **Sample (HDPE) is placed :**

- <u>at THz focus (z_{THz} = 0 mm):</u> Transmitted THz electric field through sample which is placed at the THz focus

  Files: Focus/HDPE_focus_00.dat, _01.dat, _02.dat

- <u>1 mm before THz focus (z_{THz} = -1 mm):</u> Transmitted THz electric field through sample which is placed 1 mm before the THz focus

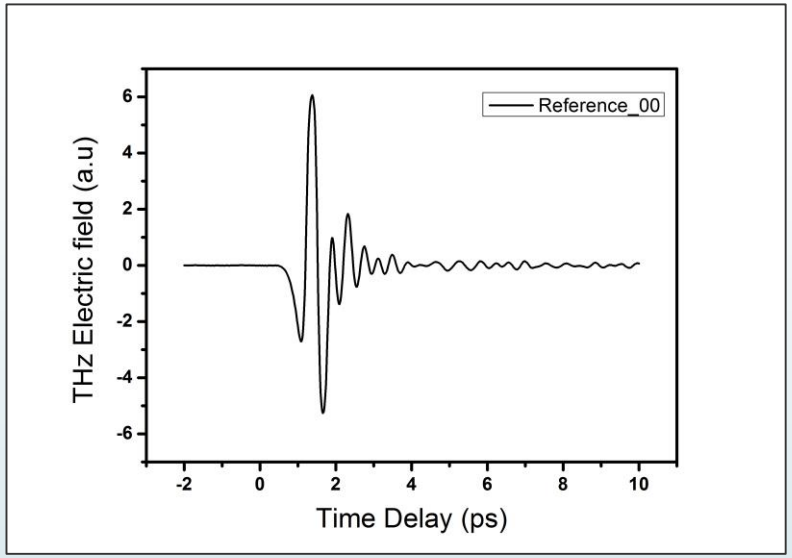
  Files: Focus_-1mm/HDPE_focus_-1mm_00.dat, _01.dat, _02.dat

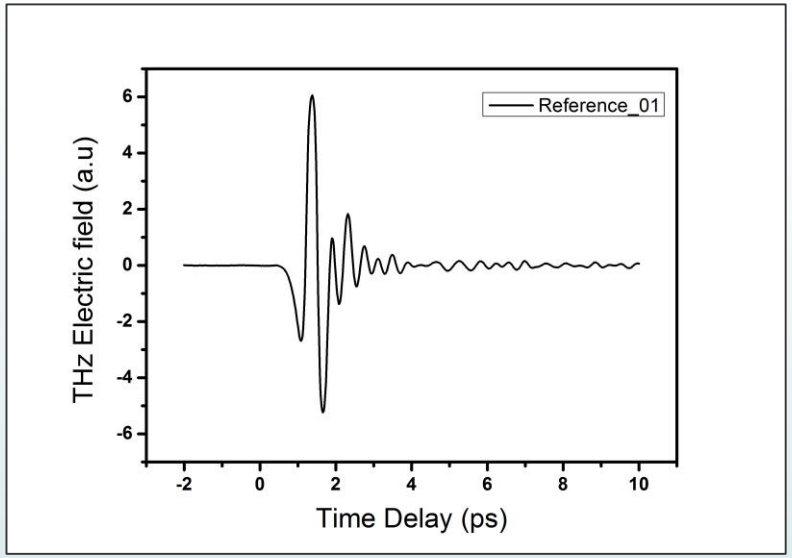
- <u>1 mm after THz focus (z_{THz} = +1 mm):</u> Transmitted THz electric field through sample which is placed 1 mm after the THz focus

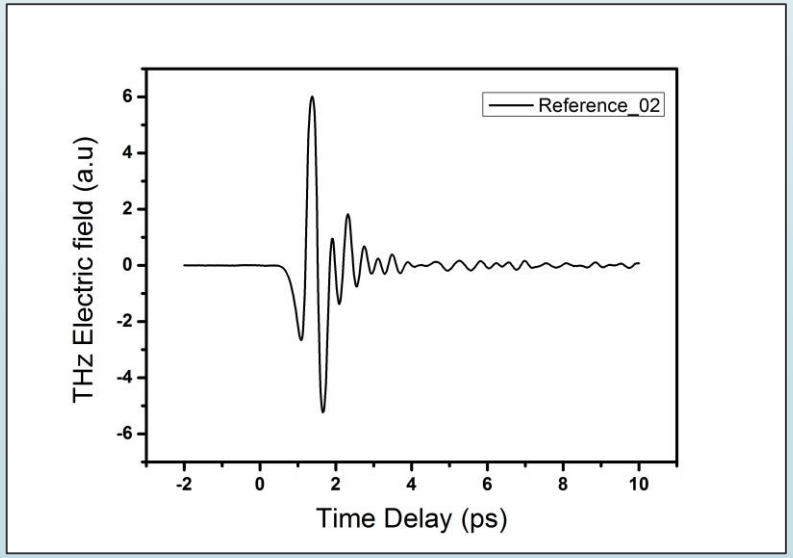
  Files: Focus_+1mm/HDPE_focus_+1mm_00.dat, _01.dat, _02.dat

Reference graphs

Reference/
Reference_00.dat
Reference_01.dat
Reference_02.dat

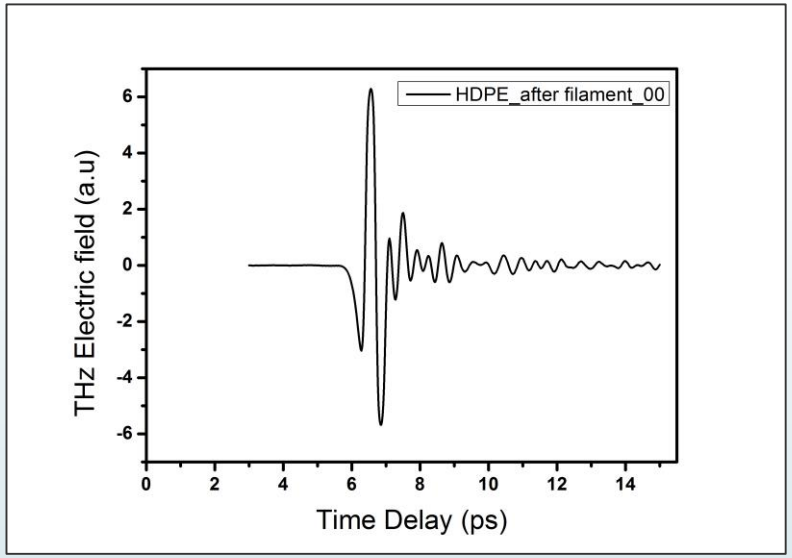


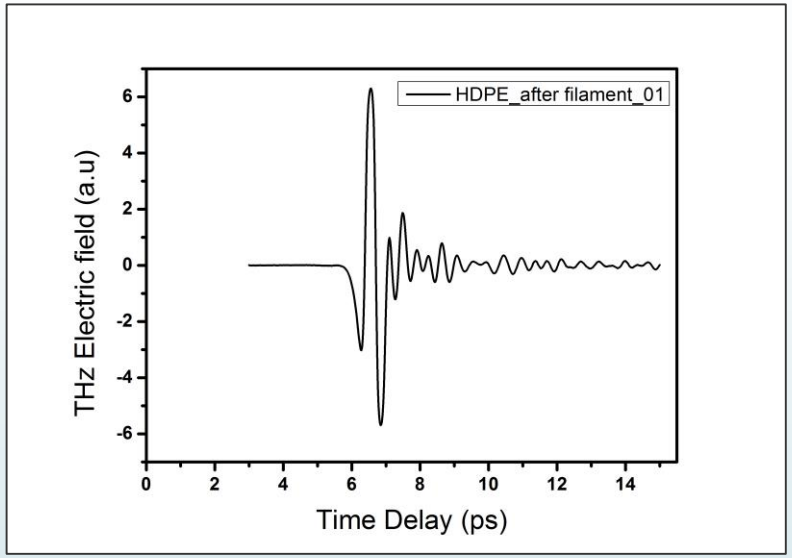


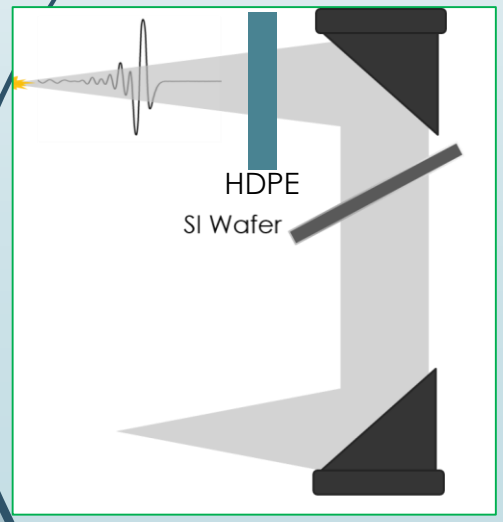


HDPE placed after the filament

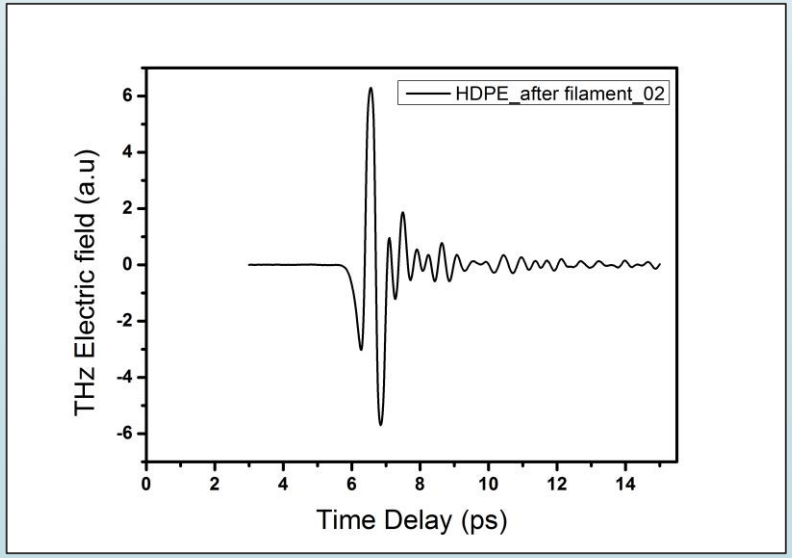
After filament/
HDPE_after_filament_00.dat
HDPE_after_filament_01.dat
HDPE_after_filament_02.dat





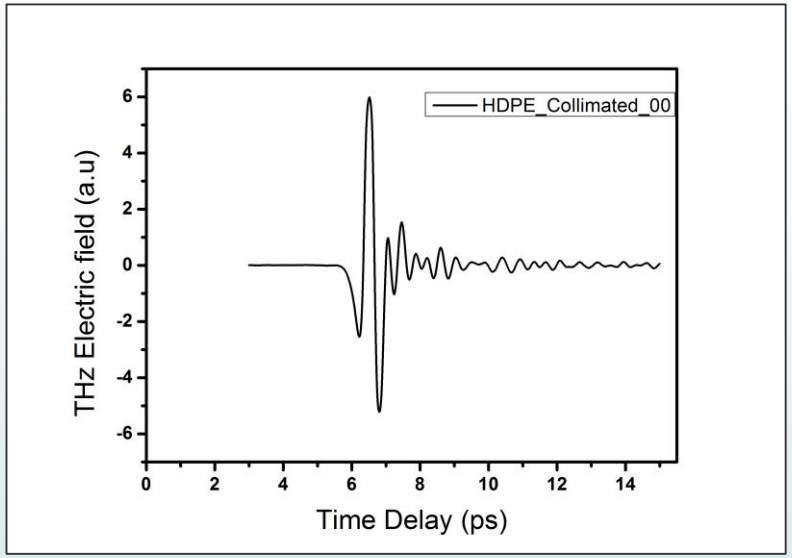


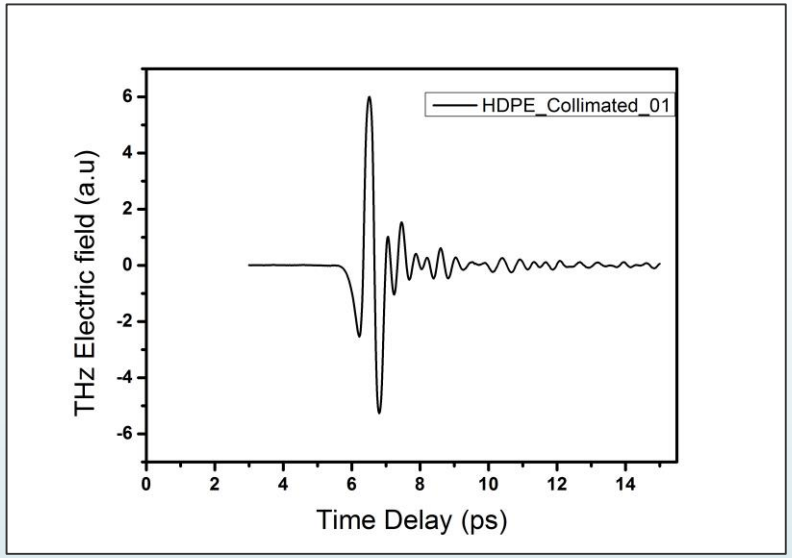
Transmitted THz electric field through the sample, which is placed after the filament, before the first parabolic mirror (P1)

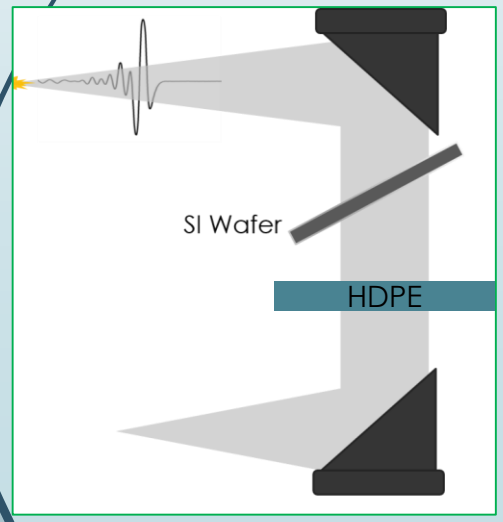


HDPE placed at collimated THz beam

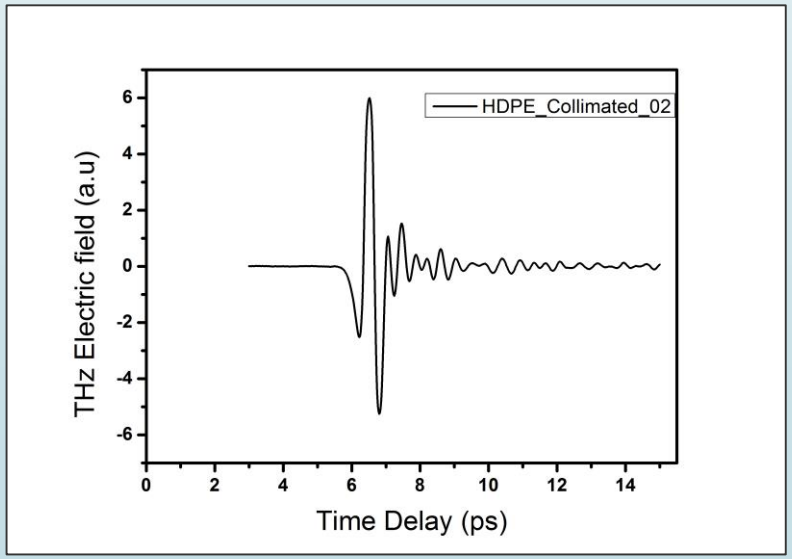
Collimated/
HDPE_collimated_00.dat
HDPE_collimated_01.dat
HDPE_collimated_02.dat





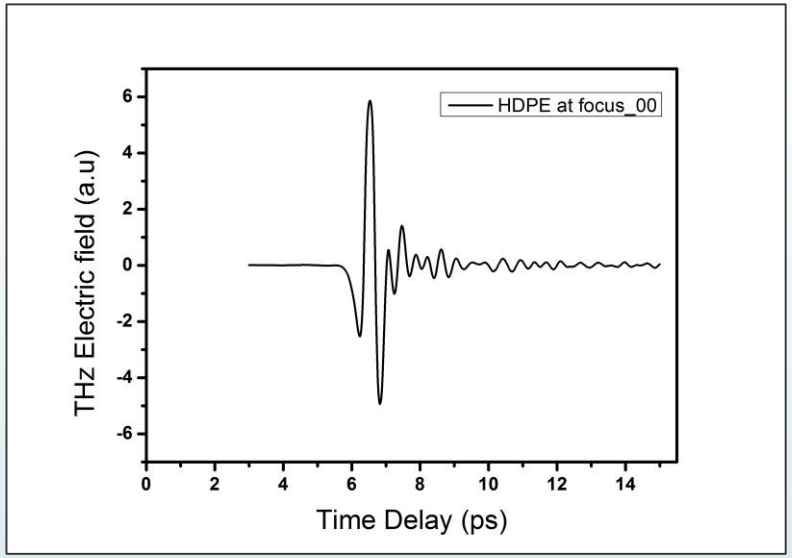


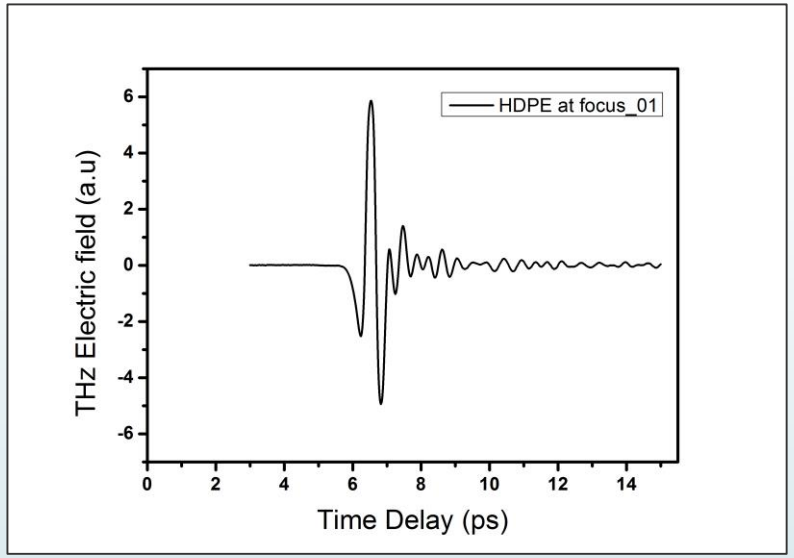
Transmitted THz electric field through the sample which is placed after the Si wafer and before the parabolic mirror (P2) where the THz beam is collimated

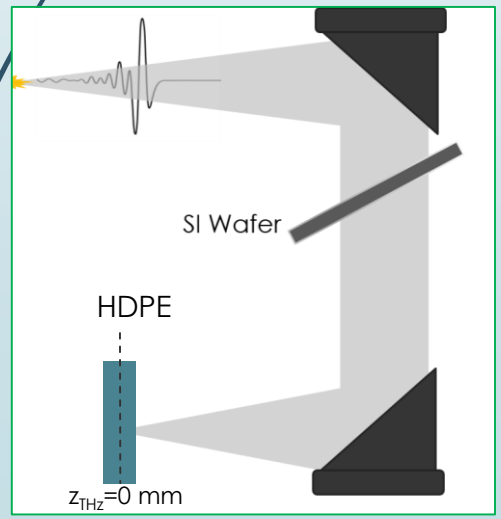


HDPE placed at THz focus (z_{THz}=0 mm)

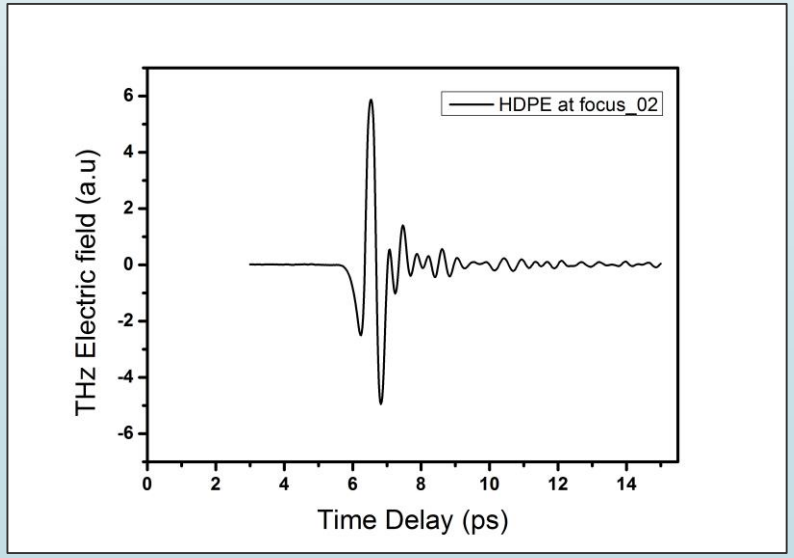
Focus/
HDPE_focus_00.dat
HDPE_focus_01.dat
HDPE_focus_02.dat





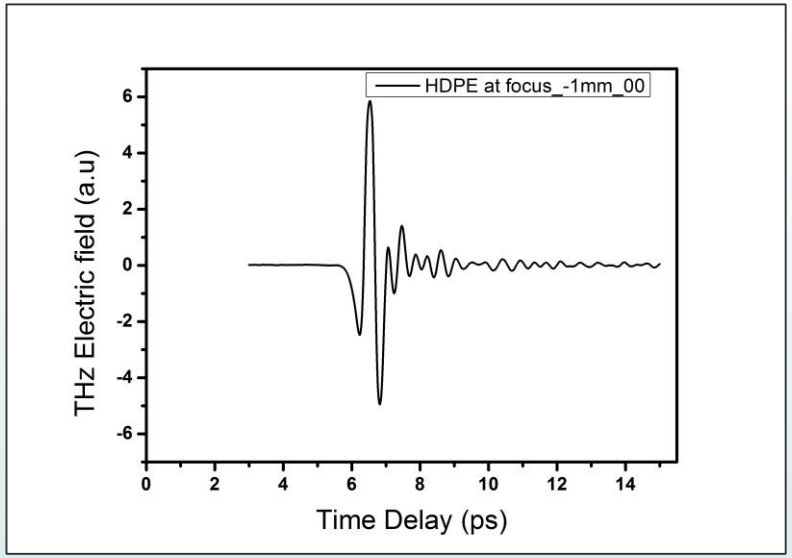


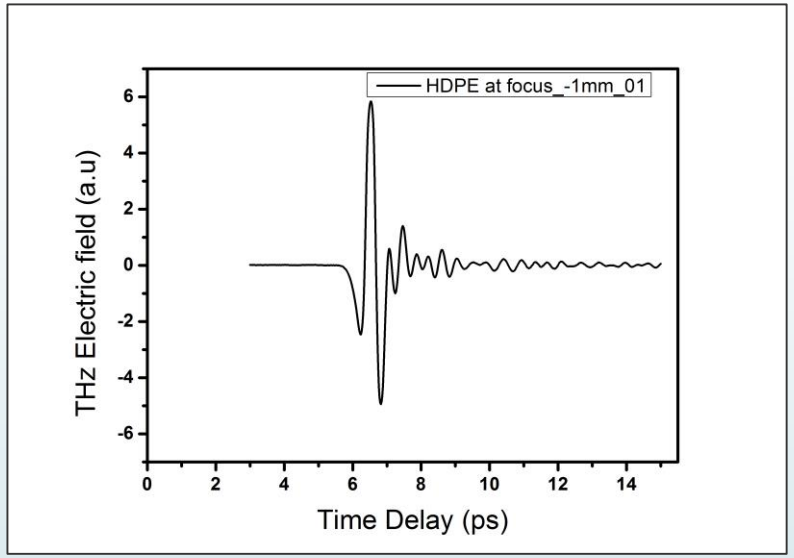
Transmitted THz electric field through sample which is placed at the THz focus

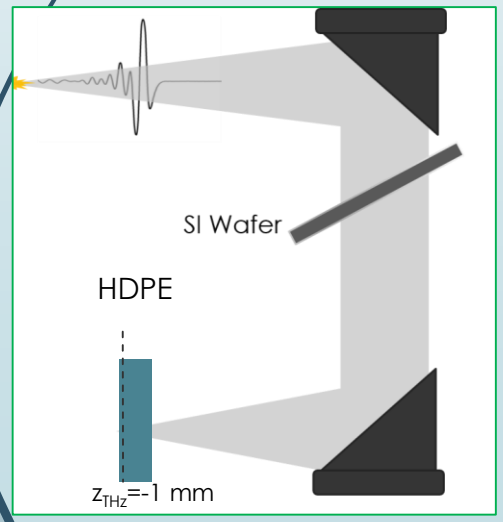


HDPE placed at z_{THz}= -1mm

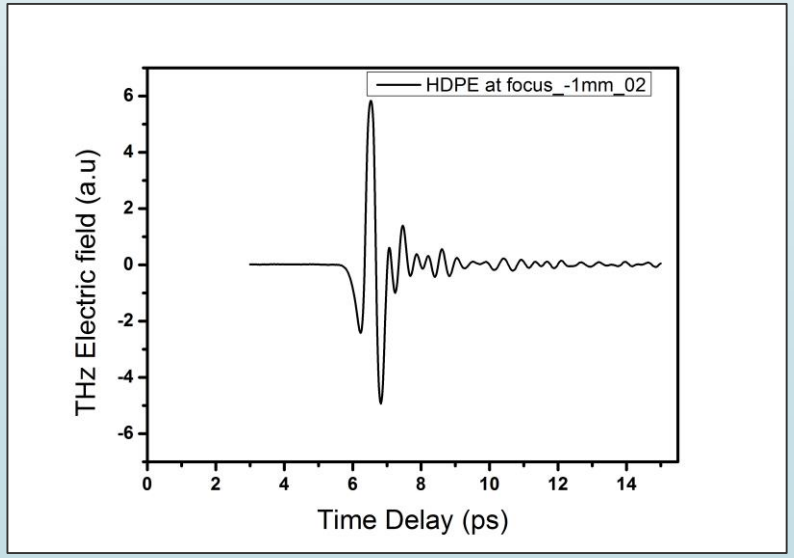
Focus_-1mm/
HDPE_focus_-1mm_00.dat
HDPE_focus_-1mm_01.dat
HDPE_focus_-1mm_02.dat





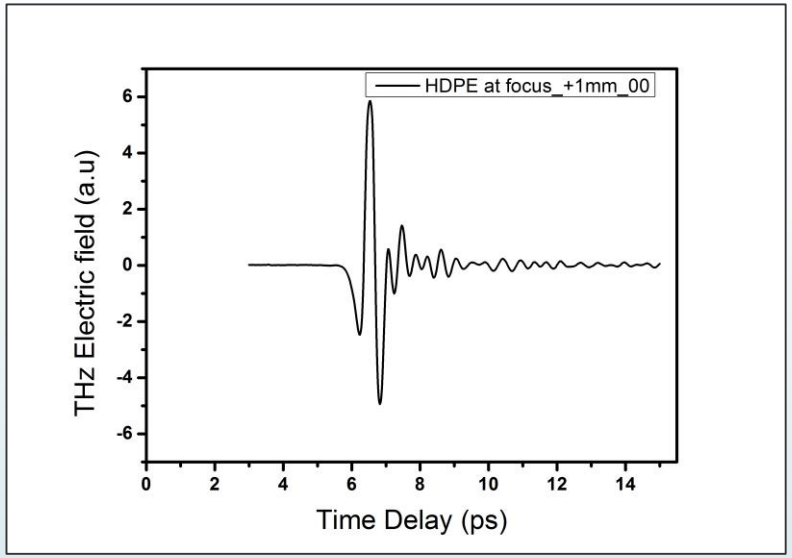


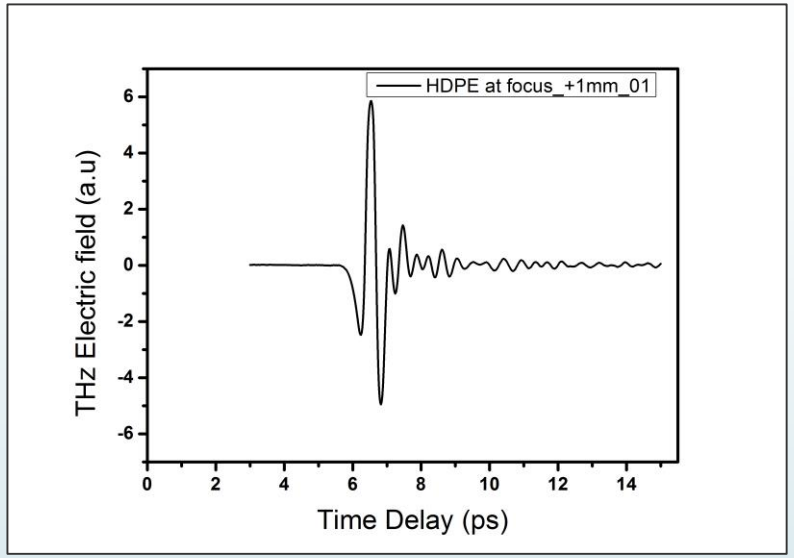
Transmitted THz electric field through sample which is placed 1 mm before the THz focus

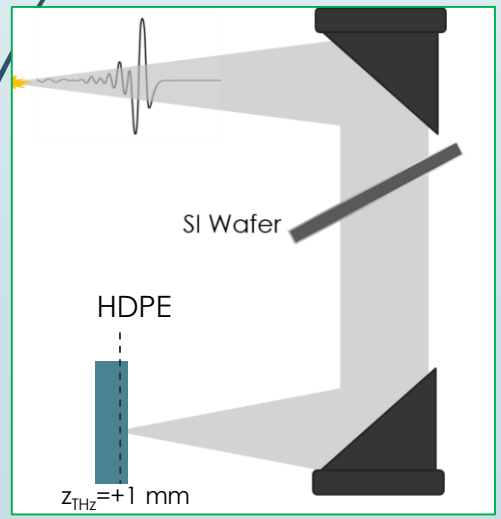
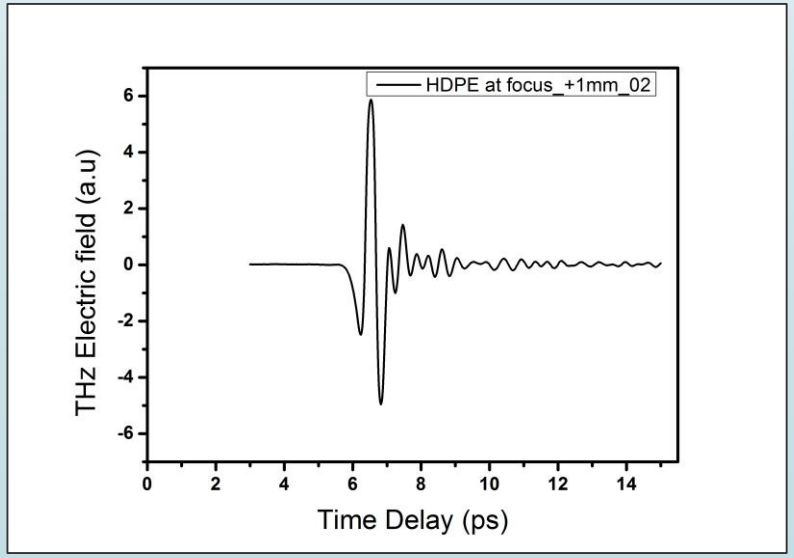


HDPE placed at z_{THz}= +1mm

Focus_+1mm/
HDPE_focus_+1mm_00.dat
HDPE_focus_+1mm_01.dat
HDPE_focus_+1mm_02.dat







Transmitted THz electric field through sample which is placed 1 mm after the THz focus